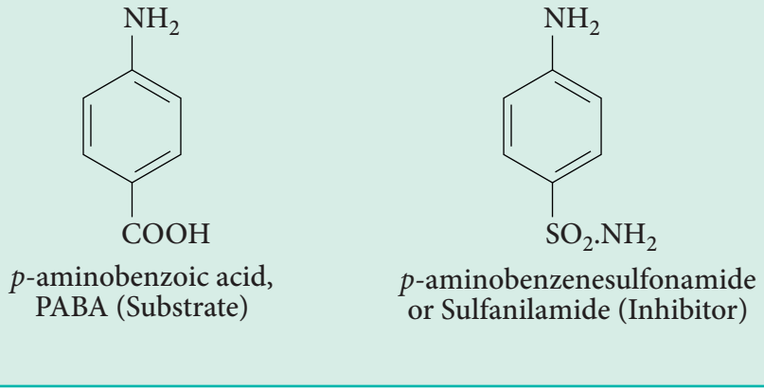
Bacteria synthesise the vitamin folic acid from P-aminobenzoicacid (PABA). Bacterial cell wall is impermeable to folic acid. Sulphanilamides and other sulpha drugs are structural analogues of PABA and hence inhibit folic acid synthesis in bacteria and they die. This drug is nontoxic to human cells, because human beings lack the enzymes necessary for folic acid synthesis and it is needed as a vitamin in the diet.



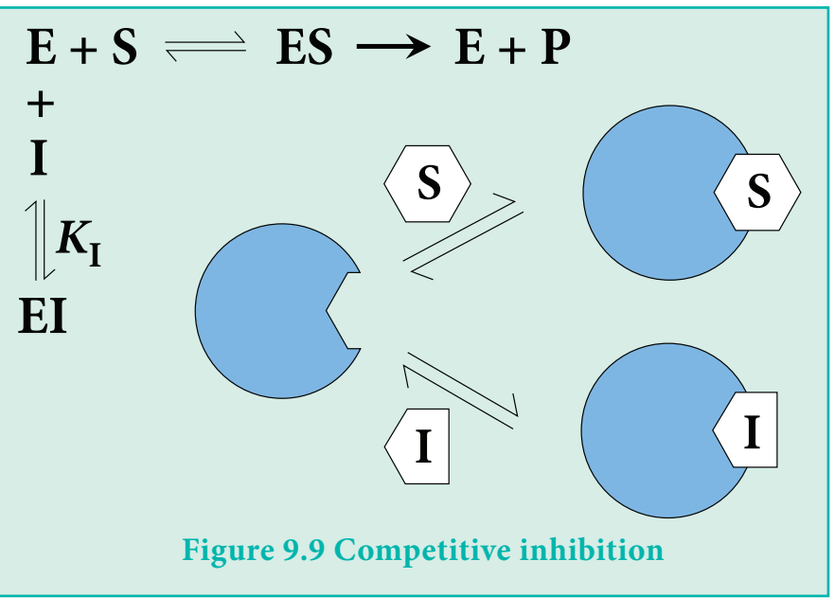




**Figure 9.9 Competitive inhibition**

The action of competitive inhibitor may be given as follows (Figure 9.9)

The inhibitor constant ($K_i$) for competitive inhibition is

$$K_i = \frac{([E]\ [I])}{([EI])}$$

### 9.3.1.2. Noncompetitive inhibiton

It is also a reversible inhibition in which the inhibitor has no structural resemblance with the substrate and binds at a site other than the active site on the enzyme surface.

In noncompetitive inhibition both the substrate and inhibitor can bind simultaneously to an enzyme molecule.

The inhibitor binds with the enzyme as well as ES complex which leads to the formation of EI and ESI complexes, respectively.

Noncompetitive inhibition cannot be overcome by increasing the substrate concentration.

In noncompetitive inhibition $K_m$ value is unchanged while $V_{max}$ is decreased.



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